



國立臺灣科技大學 電子工程系

碩士學位論文

應用於 nFAPI 分離架構之吞吐量—延遲權衡的自適
應時序控制

Adaptive Timing Control for Throughput—
Latency Trade-off in nFAPI Split

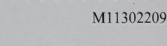
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中華民國一百一十五年六月一十五日

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摘要

在開放式無線接取網路 (O-RAN) 架構中，功能拆分選項 6 (Option 6 ，即標準化之 nFAPI) 藉由分離媒體存取控制 (MAC) 與實體 (PHY) 層，提供了在一般封包交換網路上部署的彈性，避免了時效性網路 (TSN) 硬體同步的高昂成本。然而，將 MAC 層遷移至雲端環境中的虛擬化網路功能 (VNF)，會不可避免地引入不可預期的傳輸與處理延遲。現有的開源實作多採用靜態的提前時槽 (slots ahead) 配置，無法適應動態的網路抖動，導致嚴重的排程失敗 (如遺失時槽封包) 與吞吐量下降。為解決此挑戰，本論文針對非理想的前傳 (fronthaul) 網路環境，提出了一套自適應時序控制框架。該方法採用指數平滑移動平均 (EWMA) 演算法動態估算網路延遲，並主動調整排程的提前時槽，確保控制訊息能準確落入實體網路功能 (PNF) 的嚴格時序視窗內。我們在整合商用 Pegatron O-RAN 無線單元 (O-RU) 的 OpenAirInterface (OAI) 測試平台上進行實驗驗證。結果顯示，本論文提出的自適應機制能有效消除同步失敗，維持穩定的混合自動重傳請求 (HARQ) 來回延遲 (RTT)，並在嚴重的網路壅塞下確保系統吞吐量的韌性與穩定。

Abstract

In the Open Radio Access Network (O-RAN) architecture, functional split Option 6 (standardized as nFAPI) provides deployment flexibility over general packet-switched networks by decoupling the Medium Access Control (MAC) and Physical (PHY) layers, thereby avoiding the very high costs of Time-Sensitive Networking (TSN). However, migrating the MAC layer to a Virtualized Network Function (VNF) in a cloud environment introduces non-deterministic transport and processing latency. Existing open-source implementations rely on static slots ahead timing configurations, which fail to adapt to unpredictable network jitter, leading to severe scheduling failures and throughput degradation. To address this challenge, this thesis proposes an adaptive timing control framework tailored for non-ideal fronthaul environments. The proposed method employs an Exponentially Weighted Moving Average (EWMA) to dynamically estimate network delay and actively adjust the scheduling slots ahead, ensuring that control messages arrive within the strict timing window of the Physical Network Function (PNF). Experimental validations conducted on an OpenAir-Interface (OAI) testbed integrated with a commercial Pegatron O-RAN Radio Unit (O-RU) demonstrate that the proposed adaptive mechanism effectively eliminates synchronization failures, maintains stable Hybrid Automatic Repeat Request (HARQ) round-trip times (RTT), and ensures stable system throughput under severe network congestion.

Acknowledgements

I would like to express my deepest gratitude to my advisor, Professor Ray, for his dedicated guidance and patient instruction throughout the course of my research. His mentorship has been invaluable, allowing me to gain both academic insights and practical experience that greatly enriched my personal growth.

I am also sincerely thankful to my parents, whose unwavering support enabled me to pursue my master's degree without worry and focus wholeheartedly on my studies.

I would like to thank my friend—Tzu-Yu, Mei-Chun, Yu-Hsuan, Chih-Hsuan, and Ching-Han—as well as the group “Ba Gwai.” During times of stress, they were always there to listen, have fun, and help me unwind, serving as my warmest emotional outlet. My gratitude also goes to my badminton group and NTUT basketball friends, who consistently encouraged me to exercise whenever I felt lazy—otherwise, I surely would have gained much more weight.

In the laboratory, I was fortunate to join a team with a harmonious and collaborative atmosphere. The senior members generously shared their knowledge and guided us with care, while the junior members were always willing to help. I am especially grateful for the tremendous support during my oral exam preparation. Beyond academics, we also enjoyed many group activities together, such as Christmas gift exchanges and dinners in Gongguan, which not only strengthened our bonds but also filled my graduate life with joy.

I am also very thankful for the group of wonderful friends I made in the lab—Big Winnie, Joanna, JOJO, YH, ZX, Heidi, Kenny, Ming, Toby, Richard, Mira, Will, Little Winnie, and Nino. Thank you for always being there to chat, share daily life, and accompany me through most of my master ' s journey.

Lastly, I wish to thank my classmates Jerry, Joe, Wilfrid, Johnson, Sheryl, and Sylvia. From preparing midterm exams in the first year to oral exams in the second year, the mutual support and friendship we shared have become some of my most cherished memories.



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Chapter 1 Introduction

1.1 Research Background and O-RAN Architecture Evolution

In the architectural evolution of the fifth-generation (5G) and the upcoming sixth-generation (6G) mobile communication standards, the Radio Access Network (RAN) is undergoing a fundamental transformation from the traditional monolithic base station to a highly virtualized, cloudified, and disaggregated architecture. Under the traditional architecture, baseband processing and Radio Frequency (RF) functions were tightly bound to the special hardware equipment of a single vendor, which not only limited the flexibility of network deployment but also significantly increased the capital and operational expenditures of operators.

With the rise of Cloud-RAN (C-RAN), disaggregated deployment methods have gradually become popular, centralizing the Baseband Processing Unit (BBU) into a remote BBU Pool while leaving the Remote Radio Head (RRH) at the edge site, connecting the two via the Common Public Radio Interface (CPRI). However, the C-RAN architecture is still limited by the massive fronthaul bandwidth requirements and strict latency constraints. To overcome these bottlenecks and completely break the dilemma of vendor lock-in, the O-RAN Alliance [1] and the 3rd Generation Partnership Project (3GPP) jointly promoted further disaggregation of the RAN, formally splitting the traditional BBU and RRH logically and physically into three independent network entities: the Central Unit (CU), Distributed Unit

(DU), and Radio Unit (RU) [2,3].

In this disaggregated model, functional split options have become the most critical design core for determining network performance, hardware complexity, bandwidth consumption, and latency tolerance [4,5]. 3GPP has standardized a variety of functional split options to adapt to diverse network requirements and deployment scenarios.

1.1.1 Split Option 2

Split Option 2 is a High Layer Split (HLS) architecture that primarily separates the CU and DU. Under this split, higher-layer protocols such as the Packet Data Convergence Protocol (PDCP) and Radio Resource Control (RRC) run centrally in the CU, while lower-layer Radio Link Control (RLC), Medium Access Control (MAC), and Physical Layer (PHY) remain in the DU. This architecture has relatively loose fronthaul bandwidth and latency requirements, making it easy to deploy over general packet-switched networks. The design of its synchronization mechanism is also very simple; due to its high tolerance, it only relies on flow control and deep buffering to operate stably [4], and is widely recognized as the standard configuration with the best cost-effectiveness ratio, especially suitable for non-real-time data transmission scenarios.

1.1.2 Split Option 6

Split Option 6, also known as the MAC-PHY split, precisely places the architectural separation point between the MAC layer and the PHY layer.

In this option, the DU is responsible for processing upper-layer protocols including the MAC layer, while the RU retains complete physical layer (PHY) processing capabilities. Since the fronthaul network transmits MAC Protocol Data Units (MAC PDUs) scheduled by the MAC layer, its bandwidth requirements are dynamically adjusted based on the actual load of the users. Notably, although the nFAPI-based Split Option 6 architecture also supports deployment over general packet-switched networks, its synchronization requirements are completely different from Split 2. It must strictly follow the scheduler's synchronization baseline at every slot level, ensuring data is delivered precisely on time from the VNF to the PNF. During this process, the operating system scheduling delays on the VNF side, as well as the unknown transmission delays and jitter introduced by the packet-switched network, are major challenges that must be overcome to maintain precise time synchronization [6].

1.1.3 Split Option 7.2

Split Option 7.2x (Low-PHY split) places the split point inside the physical layer, and is currently the primary Lower Layer Split (LLS) standard promoted by the O-RAN Alliance [7]. Although moving partial computations (such as FFT/IFFT) to the RU alleviates fronthaul pressure, its fronthaul transmits frequency-domain IQ symbols processed by the physical layer, demanding extremely high bandwidth. Because of this, Split 7.2 must be deployed under very strict network conditions, and its reliance on hardware equipment (such as hardware-level Precision Time Protocol, PTP) results in extremely high deployment costs [8]. However, precisely

because it has a strict private network environment and transmits a stable, continuous stream of IQ data, its synchronization mechanism is relatively stable and does not require special handling for latency variation issues caused by different offered loads.

1.1.4 Split Option 8

Split Option 8 is the most traditional lower-layer split architecture, placing the split point between the Radio Frequency (RF) and the Physical Layer (PHY). Under this architecture, all baseband processing functions (including complete PHY, MAC, and higher-layer protocols) are centralized in the DU or CU, while the RU is only responsible for RF signal transmission and reception. This configuration produces a fronthaul data stream dominated by time-domain IQ samples, usually relying on CPRI or enhanced CPRI (eCPRI) for transmission. Although it highly centralizes hardware resources, its fronthaul network bandwidth requirements are extremely high, and it is strictly constrained by system bandwidth and antenna count. Furthermore, its latency limits are extremely strict (maximum allowable one-way latency is $250\mu s$ [2]), thus, similar to Split 7.2, it heavily relies on high-quality private optical networks and precise timing synchronization hardware.

1.2 Network Architecture and Challenges Based on Split Option 6

This thesis focuses on the architectural design and application of Split Option 6. As mentioned earlier, while adopting lower-layer splits (such as Split Option 7.2) can further centralize computational resources, it requires a more expensive PTP deployment environment and a transmission network with extremely high demands for low latency, which will lead to high capital and operational expenditures. In light of this, choosing the Split Option 6 approach, which leaves the complete physical layer (Monolithic PHY) at the remote node, can strike the best balance between resource requirements and cost-effectiveness. According to research by Khan et al. [9], cost-benefit trade-off analysis for 5G NR small cells indicates that compared to strict lower-layer splits, the architecture separating the MAC and PHY layers can significantly reduce the required fronthaul bandwidth while ensuring goodput, thereby achieving the goal of lowering overall deployment costs. Therefore, in non-ideal or resource-constrained network environments, Split Option 6 becomes the ideal choice that balances performance and cost.

1.2.1 Introduction to the FAPI Interface

Within the internal architecture of 5G base stations, communication between the MAC layer and PHY layer relies on standardized interfaces. The Functional Application Platform Interface (FAPI) introduced by the Small Cell Forum (SCF) was created for this purpose [10]. FAPI provides a stan-

standardized format for control messages and data payloads, allowing MAC and PHY components from different vendors to interoperate. However, the traditional FAPI design assumes that the MAC and PHY are located within the same physical device, and the two exchange data quickly with low latency via shared memory, which limits the geographical disaggregation and distributed deployment of base station functions.

1.2.2 Introduction to the nFAPI Interface

To overcome the physical deployment limitations of FAPI and realize the Split Option 6 architecture across networks, SCF further developed the network Functional Application Platform Interface (nFAPI) [11]. The core innovation of nFAPI lies in encapsulating FAPI messages, which originally relied on shared memory, into a network packet protocol that can be transmitted over Ethernet or IP networks [12]. This technological breakthrough grants Virtual Network Functions (VNFs) immense flexibility, allowing cloudified MAC functions running on Commercial Off-The-Shelf (COTS) servers to remotely control physical PHY devices on Physical Network Functions (PNFs) over packet-switched networks.

It is particularly important to note that according to 3GPP TR 38.801 [2] specifications, the ideal fronthaul one-way latency limit for Split Option 6 is $250\mu s$. However, the SCF provides a more practical, special interpretation for nFAPI's latency tolerance in its specifications [11,13]. The SCF indicates that nFAPI is designed for packet-switched IP networks, and its fronthaul latency budget is primarily determined by the configuration of the autonomous HARQ retransmission mechanism. By configuring two

autonomous HARQ retransmissions, nFAPI can tolerate up to 6 ms of fronthaul delay (including frame delay and delay jitter) [13]. Furthermore, in a non-ideal fronthaul environment, by fine-tuning the HARQ parameters of the S-DU (such as K0, K1, K2) and the RACH timer, combined with the built-in delay management mechanism of the nFAPI P7 interface and the SCTP protocol adapted for the P5 interface, nFAPI can even support highly delayed and highly jittered transmission conditions of up to 30 ms [11]. This design specifically for non-ideal networks makes Split Option 6 highly practical and flexible in deployment.

1.2.3 Comprehensive Comparison of Split Technologies



To more intuitively present the differences between different split options, Table 1.1 provides a comprehensive comparison of Split 2, Split Option 6, Split 7.2x, and Split 8 in terms of protocol boundaries, bandwidth requirements, and latency limits.

This thesis focuses on the deployment and optimization of nFAPI because it provides excellent network flexibility, allowing network operators to logically split the base station and deploy it on two independent machines, even connected via general packet-switched networks. However, this physical split also brings immense challenges: in a non-deterministic network environment, how to maintain time synchronization between the two nodes and conduct precise delay management has become the primary difficulty in ensuring stable system operation.

Table 1.1: 5G RAN Split Deployment Comparison Table

Split Option	Split 2	Split 6 (nFAPI)	Split 7.2x	Split 8
Protocol Boundary	PDCP / RLC	MAC / PHY	Low-PHY / High-PHY	RF / PHY
Fronthaul Data Type	PDCP PDU	MAC PDU (Packetized)	Frequency-domain IQ symbols	Time-domain IQ samples
Standardized Interface	F1	SCF nFAPI	O-RAN WG4 eCPRI	CPRI / eCPRI
Bandwidth Driver	Actual user traffic	Actual user traffic	Bandwidth, spatial layers	System bandwidth, antennas
Latency Limit [2,13]	1.5 ~ 10ms	250μs (3GPP) 6 ~ 30ms (SCF)	250μs	250μs
Optimal Deployment	Non-real-time services	Small cells, indoor private networks	Macro cells	C-RAN, dedicated fiber

1.3 Research Motivation and Problem Definition

Although Split Option 6 has flexible timing adjustment capabilities and better tolerance for non-ideal fronthaul networks, it still faces severe challenges in practical deployment. As radio access networks move towards virtualization and cloudification, the hosted MAC layer functions are typically deployed as Virtual Network Functions (VNFs) on general-purpose processors. However, in this general-purpose computing environment, non-deterministic operating system scheduling jitter and processing delay uncertainty often impact timing-sensitive baseband processing [6,14].



Specifically, in existing open-source implementations (such as OpenAirInterface), the timing control of the nFAPI interface mostly relies on a statically configured slots ahead parameter. This hardcoded approach cannot dynamically adapt to the rapidly changing transmission delays and network jitter in packet-switched networks. When bursty congestion occurs in the fronthaul network or latency spikes appear in the VNF computing environment, scheduling commands and control messages are highly likely to miss their scheduled timing deadlines, thereby generating Missing Slot Packet errors. These types of errors not only lead to communication interruptions but also limit the VNF's elastic scalability in dynamic cloud resource environments.

Therefore, the core research question of this thesis is defined as follows: In a cloud environment with non-deterministic processing loads and

network jitter, how to dynamically and accurately compensate for end-to-end latency in the nFAPI Split Option 6 architecture to ensure the stability of timing synchronization?

1.4 Related Works

In recent years, the rapid development of the Open Radio Access Network (O-RAN) architecture has prompted many studies to optimize the performance and network latency management of the Split Architecture. For instance, the X5G testbed proposed by Villa et al. [8] demonstrated an advanced 5G O-RAN deployment solution based on hardware acceleration. This research successfully achieved commercial-grade peak throughput by offloading the massive computations of the Physical Layer (PHY) to dedicated NVIDIA GPUs and Mellanox SmartNICs. However, this architecture, which pursues ultimate performance, highly relies on expensive special hardware equipment, and its prerequisite is that it must operate in an ideal network environment equipped with hardware-level Precision Time Protocol (PTP) synchronization. These strict conditions greatly limit its applicability and deployment flexibility in general packet-switched networks filled with unpredictable latency and jitter.

To address the limitations of high hardware costs and strict network requirements mentioned above, this thesis proposes a highly universal software-defined solution for non-ideal fronthaul environments. Unlike X5G, which relies on hardware acceleration to overcome latency issues, and unlike traditional methods that rely on static configurations to compromise on delay jitter, this thesis designs an Adaptive Slots Ahead Control Mechanism

based on Exponentially Weighted Moving Average (EWMA) delay management specifically for the nFAPI architecture. This mechanism can run on general COTS servers, dynamically tracking and filtering network jitter, and actively adapting to unpredictable network congestion. Experimental results show that the method proposed in this thesis can effectively stabilize the HARQ loop latency (RTT) and maintain system throughput without relying on high-end special hardware or precise clock synchronization equipment, providing a more cost-effective and flexible alternative for O-RAN deployment.

Simultaneously, current academic research on O-RAN delay management has explored various dimensions to address non-deterministic latency. At the intra-node level, Lee et al. [14] proposed an adaptive transmission strategy based on the FAPI interface, utilizing the variance in Layer 1 software (L1SW) processing times to opportunistically advance transmissions, which effectively complements end-to-end transport delay management. For strict deterministic latency requirements in ultra-reliable low-latency communication (uRLLC), Adamuz-Hinojosa et al. [15] introduced the ORANUS framework, employing Stochastic Network Calculus (SNC) to provide strict mathematical bounds for latency violations. Furthermore, from the perspective of User Equipment (UE) Service Quality (QoS), Lv et al. [16] developed UQ-vRAN, which emphasizes the joint optimization of MAC scheduling and higher-layer orchestration through precise traffic digital twins.

While these existing literatures [14–17] have proposed optimization schemes ranging from specific intra-node CPU adjustments to long-term QoS orchestration, they generally do not directly address the severe cross-

node synchronization failures caused by non-deterministic fronthaul jitter in intermediate-layer splits. This thesis fills the critical gap by targeting the microsecond-level phase synchronization and latency challenges inherent in the nFAPI Split Option 6 architecture. Specifically, we focus on the fact that the Small Cell Forum (SCF) 225 nFAPI specification [11] defines the standard `Timing Info` message and interface-level delay management fields, yet existing open-source implementations only support static parameter initialization. Our proposed dynamic timing control completes this standardized framework by implementing an adaptive timing closed-loop feedback, presenting a unique academic contribution to realizing dynamic time synchronization under a pure software-defined network architecture without relying on specialized hardware.

To clearly position the proposed work within the current state-of-the-art (SOTA) literature, Table 1.2 summarizes the comparison across key architectural and performance characteristics.

1.5 Thesis Contributions

In response to the aforementioned issues, the main contributions of this thesis are as follows:

- **Multi-dimensional Latency Quantitative Analysis for nFAPI Split Option 6:** We systematically analyzed and quantified the non-deterministic factors affecting synchronization failure, including Linux kernel scheduling jitter, VNF processing load, and network queuing delays.

Table 1.2: Comparison of Proposed Work with State-of-the-Art Solutions

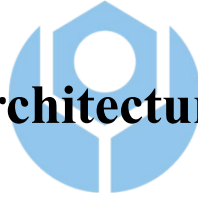
Metric / Feature	X5G [8]	Lee <i>et al.</i> [14]	Proposed Work
Split Option	Split 7.2	Split 6 (FAPI)	Split 6 (nFAPI)
Frontend Type	Dedicated Fiber / TSN	Shared Memory (Intra-node)	Packet-switched Ethernet
Hardware Dependency	GPU / SmartNIC / PTP	COTS (No Acceleration)	COTS (No Acceleration)
Control Resolution	Symbol-level	OS / Process-level	Slot / TTI-level
Adaptation Domain	Hardware Offload	Intra-node CPU Scheduling	Cross-node Jitter & Delay

- **Adaptive Slots Ahead Mechanism:** We proposed a dynamic adjustment mechanism based on Node Sync and Delay Management. It uses an EWMA model to smooth timing information to estimate real-time jitter trends, and dynamically adjusts the slots ahead (S_{ahead}) to ensure the punctuality of scheduling messages.
- **End-to-End Commercial Testbed Verification:** We implemented the proposed mechanism on the OpenAirInterface platform and combined it with a commercial UE and O-RU for verification. Experimental results show that this mechanism can effectively eliminate missing slot packet errors and maintain stable network throughput and low latency characteristics in highly congested environments.
- **Open-Source nFAPI Platform Contribution to Upstream OAI:** We submitted our adaptive timing control and nFAPI modifications as a merge request (MR) to the official OpenAirInterface (OAI) [18] repository. This contribution was merged into the official codebase. It provides a standard open-source nFAPI platform that follows the Small Cell Forum (SCF) specification for future development and testing.

Chapter 2 System Architecture

Before delving into system details, this study first introduces the adopted system framework. To integrate the network Functional Application Platform Interface (nFAPI) with mainstream commercial equipment, the system proposed in this study adopts a stable two-stage split architecture. The network is first connected through the Split 6 nFAPI interface between network functions, and then converted to the standard O-RAN Split 7.2x interface to connect to a commercial Pegatron O-RU. Figure 2.1 illustrates this system architecture and its delay management mechanism.

2.1 End-to-End Architecture



This system establishes a complete 5G end-to-end communication link extending from the core network to the user equipment:

- **Core Network (CN):** The core component responsible for routing, session management, and user data authentication.
- **O-RAN Central Unit (O-CU):** The logical node handling higher-layer protocol stacks, such as Radio Resource Control (RRC) and Packet Data Convergence Protocol (PDCP). It connects to the distributed unit via the F1 interface.
- **O-RAN Distributed Unit (O-DU):** To optimize computational performance and deployment flexibility, the system divides the O-DU into O-DU High and O-DU Low. O-DU High executes as a Virtual

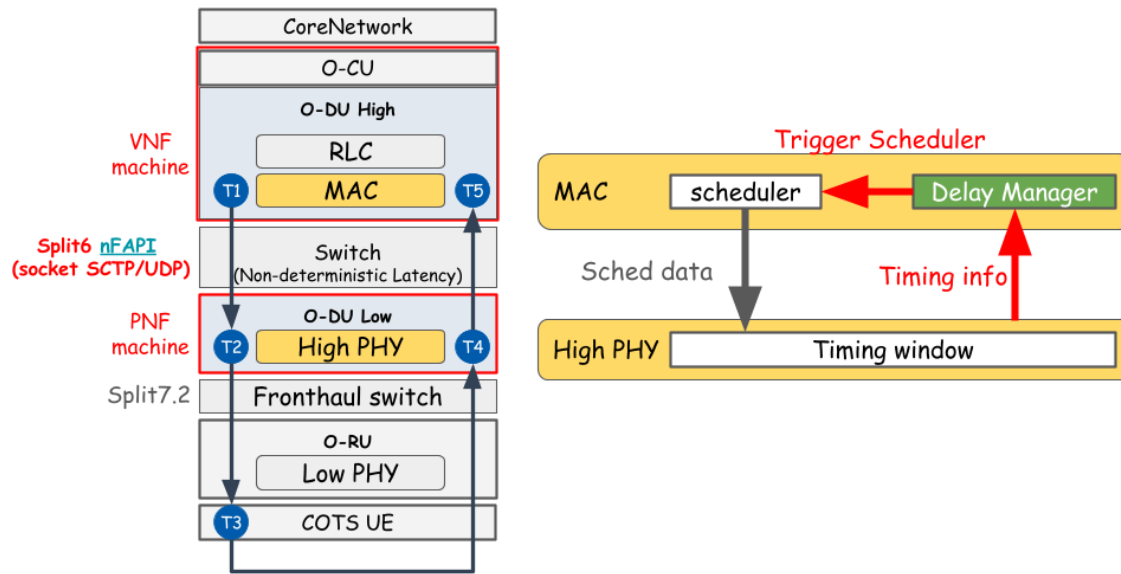


Figure 2.1: System architecture and delay management mechanism.

Network Function (VNF) on virtual machines, while O-DU Low is deployed as a Physical Network Function (PNF) on dedicated hardware.

- **Switch:** A commercial network switch located between the VNF and PNF. This intermediate layer connection utilizes the nFAPI protocol. However, traversing this packet-switched network inevitably introduces non-deterministic latency.
- **O-RAN Radio Unit (O-RU):** A dedicated hardware unit responsible for Low PHY processing, Digital Beamforming, and Radio Frequency (RF) transmission.
- **Commercial Off-The-Shelf User Equipment (COTS UE):** Standard commercial smartphones or modems acting as the network's terminal receivers.

2.2 O-DU High: MAC, Scheduler, and Timing Core

Within the O-DU High virtualized network function, the timing control is organized as a unified timing control system. The core components handling this timing core include the Scheduler, the Node Sync Module, and the nFAPI Delay Manager:

- **Scheduler:** Responsible for allocating available radio resources (e.g., time-frequency resource blocks) and determining the exact priority and timing of downlink/uplink data transmission.
- **Node Sync Module:** Responsible for software-level clock synchronization. This module periodically exchanges synchronization timestamps ($t_1 \sim t_4$) with the PNF via standard *DL/UL_node_sync* messages. It calculates the instantaneous clock offset and applies slot-level (S_{adj}) and microsecond-level (μS_{adj}) adjustments to align the VNF's local clock to the PNF baseline, eliminating long-term hardware clock drift (Clock Drift).
- **nFAPI Delay Manager:** Operates on top of the stabilized clock phase established by Node Sync. The delay manager utilizes the arrival offset (Δt_{arrive}) feedback from the PNF to dynamically estimate transport network jitter. It then actively adjusts the slots ahead (S_{ahead}) trigger offset of the scheduler. This control compensates for the non-deterministic packet transport delays and VNF processing delays without inducing synchronization loss.

2.3 O-DU Low: High PHY and Timing Window

O-DU Low manages the High PHY baseband processing. At this level, the most critical synchronization mechanism is the nFAPI PNF Timing Window:

- **Timing Window:** A predefined and strict time interval. Let T_{window} denote the duration of this PNF timing window, and T_{slot} denote the subframe slot duration. When scheduling messages (especially `DL_TTI.request`) sent by O-DU High traverse the network, they must accurately arrive and fall exactly within this window. Otherwise, the High PHY will reject the message, resulting in the inability to process data for that slot.
- **Timing Information (Δt_{arrive}):** The PNF continuously monitors the exact arrival offset of control messages relative to the transmission deadline. This offset is denoted as Δt_{arrive} , representing the remaining time margin within the timing window. The PNF periodically feeds this information back to the MAC layer delay manager. This establishes a closed-loop feedback control system to compensate for unpredictable transmission delay variations between the VNF and PNF machines.

2.4 Key Verification Metrics

To evaluate the stability, synchronization accuracy, and real-time performance of the proposed algorithm in non-ideal environments, this study defines the following key verification metrics:

- **MAC Layer HARQ RTT ($T_5 - T_1$):** Specifically measures the Hybrid Automatic Repeat Request (HARQ) loop delay at the MAC layer. This metric precisely reflects the feedback efficiency and operational health of the link layer.
- **VNF-PNF Downlink Delay ($T_2 - T_1$):** Measures the accumulated one-way downlink latency from the virtualized machine (VNF) to the physical machine (PNF). This metric is used to directly quantify and analyze the delay jitter impact introduced by the intermediate Ethernet switch.

Note: RTT stands for Round-Trip Time. To verify the stability of the split architecture under non-ideal fronthaul conditions, this study simulates network delays using Linux Traffic Control (TC) on the uplink ($T_5 - T_4$) and downlink ($T_2 - T_1$) segments.

To validate the proposed performance optimization in a realistic network environment, the system design utilizes a commercial-grade deployment framework:

- **O-RAN Split 7.2x Interoperability.**

Although the Small Cell Forum (SCF) has standardized the **network**

Functional Application Platform Interface (nFAPI) [12] for the MAC-PHY split (Option 6), commercial Radio Units (O-RUs) natively support O-RAN Split Option 7.2x. To achieve seamless integration with open-source protocol stacks like OpenAirInterface [19], this system bridges these interfaces to ensure full-scale commercial interoperability.

- **Hybrid System Architecture for End-to-End Verification.**

To verify the performance of nFAPI in a fully commercial ecosystem, this paper adopts a **hybrid architecture** (Split Option 6 + Split Option 7.2). In this model, the system provides a Translation/Interworking Layer that enables the nFAPI-driven MAC/High-PHY to successfully interact with the commercial Pegatron O-RU. This setup allows the use of **Commercial Off-The-Shelf (COTS)** user equipment to verify nFAPI performance in a real-world end-to-end environment.

Chapter 3 Proposed Method

This chapter details the proposed dynamic timing adjustment framework. To address the unstable transmission latency during packet transmission between the PNF and VNF, we propose two core timing control algorithms: Node Sync for software-level clock synchronization and Delay Management for slot-level adaptive delay management. The timing control of this system is organized as a closed-loop system, ensuring that the VNF-side scheduling can dynamically track and compensate for both microsecond-level clock drift and slot-level transport latency variation. The two core algorithms cooperate as follows:

- 
1. **Node Sync:** Maintains phase and slot boundary alignment between the VNF and PNF. It periodically exchanges timestamps to calculate the absolute phase offset, correcting for hardware clock drift (Clock Drift).
 2. **Delay Management:** Tracks and estimates packet transport delays and jitter at the slot level, dynamically adjusting the slots ahead (S_{ahead}) to guarantee that control messages fall within the PNF's strict timing window.

3.1 Node Sync

To perform software-level clock synchronization, the system adopts a timestamp exchange mechanism. The VNF periodically transmits a down-

link node synchronization message with a local timestamp t_1 . Upon receiving the message, the PNF records the local arrival timestamp t_2 and transmits an uplink node synchronization message back to the VNF, including the transmit timestamp t_3 and the echoed t_2 . When the VNF receives the message, it records its local arrival timestamp t_4 . Because the nFAP timestamps are constrained by a 1024 subframe number (SFN) loop, they wrap around every 10.24 seconds (10240000 μ s). The algorithm calculates the differences $d_1 = t_2 - t_1$ and $d_2 = t_4 - t_3$, applying a wrap-around correction to align them within the interval $[-5120000, 5120000]$ μ s. The VNF then calculates the instantaneous clock offset:

$$offset = \frac{d_1 - d_2}{2} \quad (3.1)$$

To prevent timing adjustments from causing control loop oscillations, the synchronization control loop implements a state-based locking mechanism. The system has two synchronization states: locked ($sync_locked = 1$) and unlocked ($sync_locked = 0$). In the locked state, the VNF monitors the drift of the local clock. If the absolute offset exceeds the slot duration T_{slot} :

$$|offset| > T_{slot} \quad (3.2)$$

the VNF unlocks the synchronization ($sync_locked = 0$) to trigger re-synchronization. In the unlocked state, the VNF checks whether the offset has converged within the slot duration T_{slot} . If $|offset| \leq T_{slot}$, the system locks the synchronization state ($sync_locked = 1$). Otherwise, the VNF decomposes the offset into a slot-level adjustment S_{adj} and a microsecond-level adjustment μS_{adj} based on the slot duration T_{slot} :

$$S_{adj} = offset / T_{slot} \quad (3.3)$$

$$\mu S_{adj} = offset \pmod{T_{slot}} \quad (3.4)$$

These adjustments are accumulated into the global adjustment variables:

$$slot_adjustment \leftarrow slot_adjustment + S_{adj} \quad (3.5)$$

$$us_adjustment \leftarrow us_adjustment - \mu S_{adj} \quad (3.6)$$

The negative sign in the microsecond adjustment reduces the local thread sleep time when the VNF clock is behind, speeding up the slot transition rate to catch up. The synchronization process is summarized in Algorithm 1.



Algorithm 1 Node Sync

Input: t_1, t_2, t_3 (Timestamps from PNF *UL_node_sync*), t_4 (VNF local receive timestamp)

Output: *slot_adjustment* (Slot adjustment accumulator), *us_adjustment* (Microsecond adjustment accumulator)

Notation:

offset (Instantaneous timing offset), *sync_locked* (Synchronization lock state)

T_{slot} (Slot duration in μs), T_{wrap} (Wrap-around period, 10240000 μs)

WrapCorrection(x, T) (Corrects x within $[-T/2, T/2]$)

Phase I: Timing Offset Calculation

- 1: $d_1 \leftarrow \text{WrapCorrection}(t_2 - t_1, T_{\text{wrap}})$
- 2: $d_2 \leftarrow \text{WrapCorrection}(t_4 - t_3, T_{\text{wrap}})$
- 3: $\text{offset} \leftarrow (d_1 - d_2)/2$

Phase II: Hysteresis Lock and Clock Adjustment

- 4: **if** $\text{sync_locked} = 1 \wedge |\text{offset}| > T_{\text{slot}}$ **then**
 - 5: $\text{sync_locked} \leftarrow 0$ ▷ Unlock due to excessive clock drift
 - 6: **end if**
 - 7: **if** $\text{sync_locked} = 0$ **then**
 - 8: **if** $|\text{offset}| \leq T_{\text{slot}}$ **then**
 - 9: $\text{sync_locked} \leftarrow 1$ ▷ Lock synchronization
 - 10: **else**
 - 11: $S_{\text{adj}} \leftarrow \text{offset}/T_{\text{slot}}$ ▷ Integer division
 - 12: $\mu S_{\text{adj}} \leftarrow \text{offset} \pmod{T_{\text{slot}}}$
 - 13: $\text{slot_adjustment} \leftarrow \text{slot_adjustment} + S_{\text{adj}}$
 - 14: $\text{us_adjustment} \leftarrow \text{us_adjustment} - \mu S_{\text{adj}}$
 - 15: **end if**
 - 16: **end if**
 - 17: **return** $\text{slot_adjustment}, \text{us_adjustment}$
-

3.2 Delay Management

To counteract changing network latency and maintain scheduling stability, a dynamic delay management strategy is implemented. This strategy operates as an Asymmetric Feedback Controller based on a Risk-Aware Predictive Model, periodically evaluating the observed worst-case late arrival offset (e_{late}) reported by the PNF.

3.2.1 EWMA-Based Risk and Uncertainty Estimation

To smooth high-frequency network noise while remaining sensitive to timing drift, the VNF Delay Manager maintains a running mean of the late offset (μ_{late}) and the network jitter (v_{jitter}). Let $e_{\text{late},i}$ be the observed worst-case late offset in scheduling slot i . The EWMA updates for mean latency and jitter are defined as:

$$\mu_{\text{late},i} = \mu_{\text{late},i-1} + \alpha \cdot (e_{\text{late},i} - \mu_{\text{late},i-1}) \quad (3.7)$$

$$v_{\text{jitter},i} = v_{\text{jitter},i-1} + \beta \cdot (|e_{\text{late},i} - \mu_{\text{late},i}| - v_{\text{jitter},i-1}) \quad (3.8)$$

where the smoothing factors are chosen as $\alpha = 1/8$ to track the long-term latency trend, and $\beta = 1/4$ to capture fast jitter fluctuations. These parameters are designed as negative powers of two ($\alpha = 2^{-3}$, $\beta = 2^{-2}$) to allow the controller to perform fast bitwise shift operations in real time.

To guide timing adjustments, the controller combines the mean late offset and the estimated jitter to predict the timing risk (R_{pred}). The predicted timing risk for slot i is calculated as the sum of the worst-case boundary of

the late offset and the network jitter:

$$R_{\text{pred},i} = \max(e_{\text{late},i}, \mu_{\text{late},i}) + v_{\text{jitter},i} \quad (3.9)$$

Based on this risk estimation, the controller recursively tracks the risk debt (D_{risk}) and the safety margin (M_{safe}):

$$D_{\text{risk},i} = D_{\text{risk},i-1} + \alpha \cdot (\max(0, R_{\text{pred},i}) - D_{\text{risk},i-1}) \quad (3.10)$$

$$M_{\text{safe},i} = M_{\text{safe},i-1} + \alpha \cdot (\max(0, -R_{\text{pred},i}) - M_{\text{safe},i-1}) \quad (3.11)$$

The risk debt D_{risk} aggregates timing violations and predicted risks over time, and the safety margin M_{safe} tracks the average buffer capacity when the system operates without risk. The system is considered debt-free if D_{risk} decays to zero ($D_{\text{risk}} \approx 0$).



3.2.2 Asymmetric Predictive Feedback Control

The update of the slots ahead trigger timing S_{next} is governed by an asymmetric control law. Under this law, the scheduler reacts immediately to timing violations or positive risk, while slowly decreasing the lead time only when the system has remained stable for a sustained duration:

- **Rapid Compensation (Preserve Throughput):** If a timing violation is detected ($e_{\text{late}} > 0$) or the predicted risk is positive ($R_{\text{pred}} > 0$), the controller immediately increments the slots ahead parameter to prevent scheduling timeouts:

$$S_{\text{next}} = \min \left(S_{\text{curr}} + 1, \left\lfloor \frac{T_{\text{window}}}{T_{\text{slot}}} \right\rfloor \right) \quad (3.12)$$

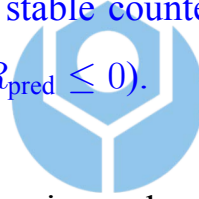
and resets the stable observation counter $C_{\text{stable}} = 0$.

- **Gradual Relaxation (Optimize Latency):** If the system is cleared of risk debt ($D_{\text{risk}} = 0$), the current sample has no risk ($R_{\text{pred}} \leq 0$), and the safety margin is sufficient ($M_{\text{safe}} \geq v_{\text{jitter}}$), the controller decrements the slots ahead timing once the stability condition is satisfied:

$$S_{\text{next}} = \max(S_{\text{curr}} - 1, 1) \quad (3.13)$$

and resets $C_{\text{stable}} = 0$. To prevent premature timing reductions as the scheduling buffer grows, the required stable count threshold increases quadratically with S_{curr} .

- **Maintain Baseline:** Otherwise, the controller holds the current timing, $S_{\text{next}} = S_{\text{curr}}$. The stable counter C_{stable} is incremented if the current sample is safe ($R_{\text{pred}} \leq 0$).



The core adjustment strategy is condensed in Algorithm 2.

3.3 Coordination between Node Sync and Delay Management

To ensure stable operation under real-world conditions, Node Sync and Delay Management must coordinate without inducing control loop oscillations. The microsecond-level Node Sync focuses strictly on correcting clock phase offsets to align the VNF's scheduling thread with the PNF's slot transitions. Because it directly shifts the VNF's time base, any clock adjustment ($S_{\text{adj}}, \mu S_{\text{adj}}$) temporarily shifts the arrival timestamp reference. The Delay Manager is designed to absorb this transient shifting. By rely-

Algorithm 2 Delay Management

Input: e_{late} (Worst-case observed late offset), S_{curr} (Current trigger timing),
 T_{window} (Timing window), T_{slot} (Slot duration)

Output: S_{next} (Trigger timing for the next MAC schedule)

Notation:

μ_{late} (EWMA mean late offset), v_{jitter} (EWMA network jitter)

R_{pred} (Predicted timing risk), D_{risk} (EWMA risk debt), M_{safe} (EWMA safety margin)

C_{stable} (Stable observation counter)

α, β (Smoothing factors for latency and jitter)

Phase I: EWMA Latency and Jitter Estimation

- 1: $\mu_{\text{late}} \leftarrow \mu_{\text{late}} + \alpha \cdot (e_{\text{late}} - \mu_{\text{late}})$ ▷ Update mean late offset
- 2: $v_{\text{jitter}} \leftarrow v_{\text{jitter}} + \beta \cdot (|e_{\text{late}} - \mu_{\text{late}}| - v_{\text{jitter}})$ ▷ Update network jitter
- 3: $R_{\text{pred}} \leftarrow \max(e_{\text{late}}, \mu_{\text{late}}) + v_{\text{jitter}}$ ▷ Evaluate predicted risk
- 4: $D_{\text{risk}} \leftarrow D_{\text{risk}} + \alpha \cdot (\max(0, R_{\text{pred}}) - D_{\text{risk}})$ ▷ Update risk debt
- 5: $M_{\text{safe}} \leftarrow M_{\text{safe}} + \alpha \cdot (\max(0, -R_{\text{pred}}) - M_{\text{safe}})$ ▷ Update safety margin

Phase II: Asymmetric Timing Adjustment

- 6: **if** $e_{\text{late}} > 0 \vee R_{\text{pred}} > 0$ **then** ▷ Rapid Compensation
 - 7: $S_{\text{next}} \leftarrow \min(S_{\text{curr}} + 1, \lfloor T_{\text{window}} / T_{\text{slot}} \rfloor)$
 - 8: $C_{\text{stable}} \leftarrow 0$
 - 9: **else if** $D_{\text{risk}} \approx 0 \wedge R_{\text{pred}} \leq 0 \wedge M_{\text{safe}} \geq v_{\text{jitter}}$ **then**
 - 10: **if** stability condition is satisfied **then** ▷ Gradual Relaxation
 - 11: $S_{\text{next}} \leftarrow \max(S_{\text{curr}} - 1, 1)$
 - 12: $C_{\text{stable}} \leftarrow 0$
 - 13: **else**
 - 14: $S_{\text{next}} \leftarrow S_{\text{curr}}$
 - 15: $C_{\text{stable}} \leftarrow C_{\text{stable}} + 1$
 - 16: **end if**
 - 17: **else**
 - 18: $S_{\text{next}} \leftarrow S_{\text{curr}}$
 - 19: $C_{\text{stable}} \leftarrow 0$
 - 20: **end if**
 - 21: **return** S_{next}
-

ing on the history-weighted risk debt (D_{risk}), the Delay Manager avoids reacting to the temporary timing offsets caused by clock adjustments, allowing Node Sync to converge. Once Node Sync locks synchronization ($sync_locked = 1$), the time base stabilizes, allowing the Delay Manager to achieve precise slot-level optimizations (S_{ahead}).



Chapter 4 Experimental Results

This study conducts experiments in an OpenAirInterface (OAI) nFAPI end-to-end environment to verify the proposed delay management method. To ensure the statistical reliability of the evaluation, each throughput value reported represents the average of 180 data points (3 independent trials $\times$ 60 samples per trial).

To validate the hypothesis and the effectiveness of the proposed dynamic trigger scheduling algorithm, the experimental scenarios are arranged as follows: Section 4.1 outlines the hardware and software environment as well as the automated evaluation framework. Section 4.2 validates the Exponentially Weighted Moving Average (EWMA) mechanism for the PNF arrival offset (Δt_{arrive}). Section 4.3 benchmarks the performance of peak throughput and MAC layer Round-Trip Time (RTT) against static baselines. Section 4.4 evaluates the stability of the proposed method under varying degrees of network congestion.

4.1 Experimental Scenario

The testbed environment connects a series of high-performance servers, radio units, and commercial smartphones. The detailed hardware and software configuration of the end-to-end network deployment is shown in the table below.

To ensure consistent data quality for automated measurement and analysis, this study developed an automated evaluation rApp running on the

Table 4.1: Testbed Node Configuration

Unit	Specification
Core Network (CN)	Open5GS [20]
Software Protocol Stack / Branch	OpenAirInterface [21] (2026.w13)
VNF Server	Intel® Xeon® Gold 6226R [22] with NetXtreme BCM5719 NIC [23,24]
PNF Server	Intel® Xeon® Gold 6433N [25] with Intel I210 NIC [24,26]
O-RU	Pegatron RU (P5G-Indoor O-RU-PR1450)
Commercial Off-The-Shelf (COTS) UE	Samsung Galaxy S20 [27]

Table 4.2: Wireless System Parameters

Parameter	Setting
Subcarrier Spacing	30 kHz
TDD Pattern	3D1S1U
MIMO / Ports	4-layer / 4T4R
Fronthaul Interface	O-RAN F release

Non-Real-Time RAN Intelligent Controller (Non-RT RIC) within the O-RAN Service Management and Orchestration (SMO) framework [28,29], as shown in Figure 4.1. This rApp automatically starts the VNF and PNF. Once the gNB is successfully established, the rApp automatically connects to the control PC via SSH, toggles the UE’s airplane mode using the Android Debug Bridge (ADB) [30], and then launches an iPerf [31] server on the Core Network server via SSH. Subsequently, it automatically executes a series of predefined iPerf test cases. After that, the rApp collects all log files via O1 SFTP and uses Python scripts to automatically plot charts for subsequent data analysis. During the experiments, we also used tcpdump [32] for packet sniffing, and utilized Linux PTP [33] and TSN time synchronization mechanisms [34] to ensure baseline timing among network nodes, while underlying data transmission relied on DPDK [35] for packet acceleration. A key capability of this framework is the ability to automate network delay simulations on the MAC feedback link ($T_5 - T_4$) and fronthaul link ($T_2 - T_1$) using Linux Traffic Control (TC), which is crucial for verifying the stability of the disaggregated architecture under non-deterministic latency.

4.1.1 Rigorous System-Level Validation and Generalization

To prove the stability, generalization capability, and compatibility of the proposed timing control and nFAPI modifications, our software implementation was submitted and successfully merged into the official upstream OpenAirInterface (OAI) repository [36]. Beyond our localized end-

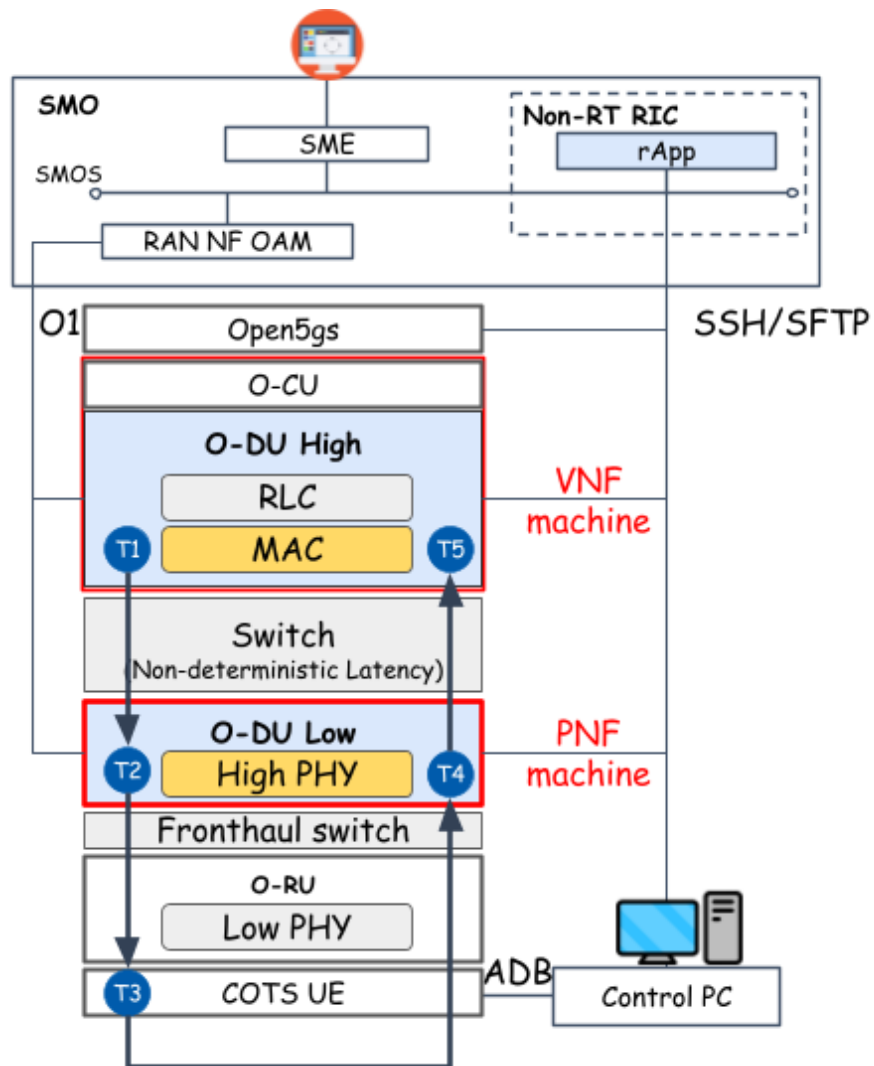


Figure 4.1: Architecture of the rApp automated evaluation framework.

to-end evaluation using the rApp framework, the proposed solution underwent rigorous testing in the official OAI Continuous Integration and Continuous Deployment (CI/CD) pipeline [18]. This automated pipeline executes multi-platform builds and physical layer regression suites to ensure production-grade software quality.

In the software compilation and image deployment stage, the proposed implementation demonstrates high portability across heterogeneous hardware architectures. As compiled in Table 4.3, the cross-platform builds successfully passed on standard x86 servers, ARM64-based edge servers, and NVIDIA Jetson edge AI platforms under both Ubuntu and enterprise-grade Red Hat Enterprise Linux (RHEL) [37] environments. These results confirm that our proposed timing control is independent of specific hardware acceleration cards, allowing seamless deployment on general-purpose Commercial Off-The-Shelf (COTS) cloud infrastructure.

Following cross-platform compilation, our implementation was subjected to system integration and physical-layer dynamic testing. Table 4.4 summarizes the key integration and simulation tests that our code passed. The validation demonstrates that the slot-level scheduler adjustments (S_{ahead}) and microsecond-level clock corrections ($S_{adj}, \mu S_{adj}$) do not break 3GPP-compliant physical layer procedures [38] or F1/N2 standard interface control-plane handovers [39]. Crucially, the code successfully integrated with commercial Quectel modules under standard RF conditions, proving that the proposed software-defined timing loops successfully satisfy the strict timing window constraints of real-world physical basebands.

These regression and compilation results prove that the proposed dy-

Table 4.3: OAI CI/CD Cross-Platform Compilation and Image Building Results

Stage	Job / Builder	Target and Hardware Description	Result
x86 Build	Ubuntu-Image-Builder	Build OAI network function images on standard x86 servers	Passed
ARM Build	Ubuntu-ARM-Image-Builder	Build images on ARM64 for ARM-based edge servers	Passed
Edge AI Build	Ubuntu-Jetson-Image-Builder	Build light-weight images for NVIDIA Jetson platforms	Passed
RHEL Build	RHEL-Cluster-Image-Builder	Build images for Red Hat Enterprise Linux (RHEL) clusters	Passed
Cross-Compile	ARM-Cross-Compile	Verify cross-compilation from x86 to ARM64 architecture	Passed

dynamic delay-aware timing control is robust, backward-compatible, and fully standardized, serving as a stable baseline for future O-RAN Option 6 deployments in the global open-source community.

4.2 Scenario I: EWMA Validation for PNF Arrival Offset (Δt_{arrive})

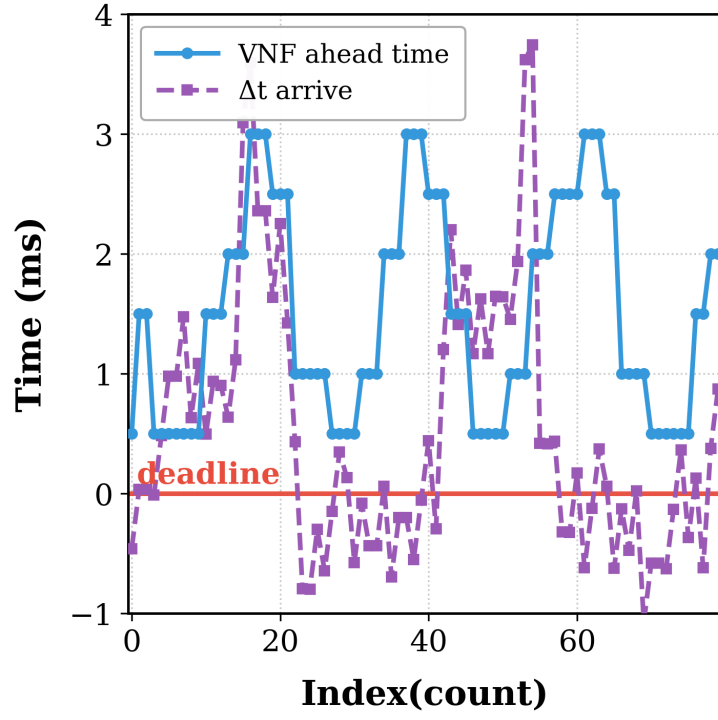
This scenario verifies the effectiveness of applying the Exponentially Weighted Moving Average (EWMA) to smooth the PNF arrival offset (Δt_{arrive}). Due to network jitter and processing delays, the raw timing offset received from the PNF can exhibit high variability. Figure 4.2 illustrates the impact of EWMA processing. Without EWMA (Figure 4.2a), the arrival offset

Table 4.4: Summary of System Integration and Physical Layer Validation

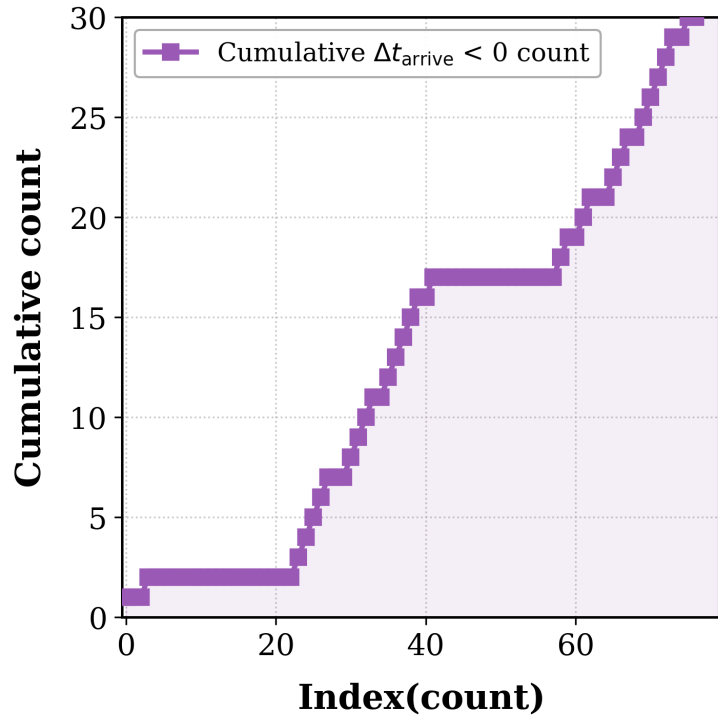
Validation Scope	Passed Job Name	Description and Highlights
E2E 5G SA and RIC Integration	OAI-FLEXRIC-RAN-Integration	Run in 5G SA mode, integrate with 5G Core and FlexRIC (near-RT RIC) [40] with low packet loss. This validation uses the E2 interface protocol [41].
Commercial UE and RF Integration	RAN-NSA-B200-Module-LTEBOX RAN-SA-B200-Module-SABOX	Run with USRP B200 and commercial Quectel modules to meet strict timing in real RF environments.
Third-Party RF Frontend Compatibility Handover Stability	RAN-SA-AW2S-CN5G RAN-SA-Handover-CN5G	Run end-to-end tests with third-party AW2S RF equipment (20MHz TDD SA) [42] and Amarisoft UE. Pass F1-interface and N2-interface handover tests [39] under 5G SA to keep control plane stable.
5G L1 and Channel Compatibility	RAN-Channel-Simulation RAN-PhySim-GraceHopper-5G	Pass physical layer simulations (including LDPC, Polar, PBCH, MIMO, MCS) to prove timing changes do not break 3GPP rules [38]. Testing was also performed on GPU-accelerated platforms [43].

fluctuates drastically, which can lead to unstable scheduling decisions. By applying EWMA (Figure 4.2b), high-frequency jitter is effectively filtered, providing a stable baseline for the timing adjustment algorithm. Furthermore, as shown in Figures 4.2c and 4.2d, the smoothed data allows the system to make relatively stable judgments. When Δt_{arrive} begins to show a downward trend, the algorithm steadily identifies this trend and actively advances more time (S_{ahead}) to maintain synchronization stability.



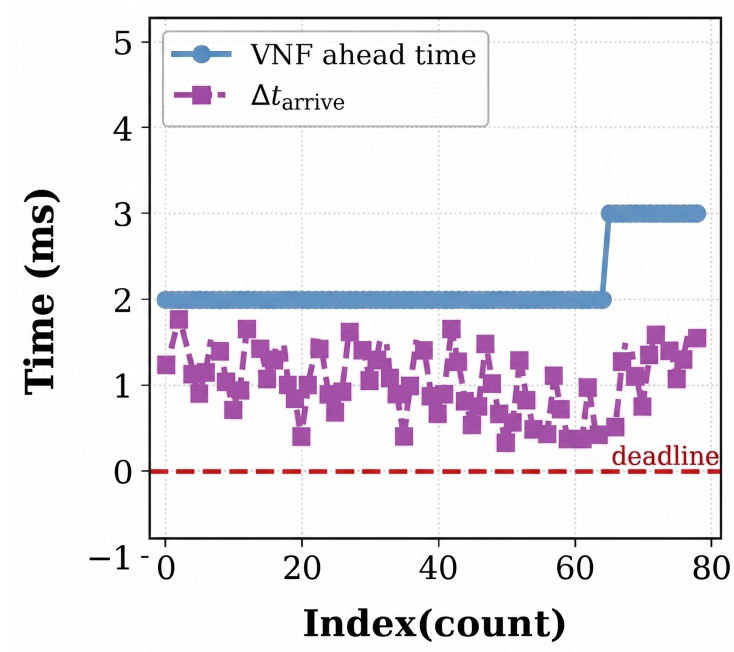


(a) Without EWMA

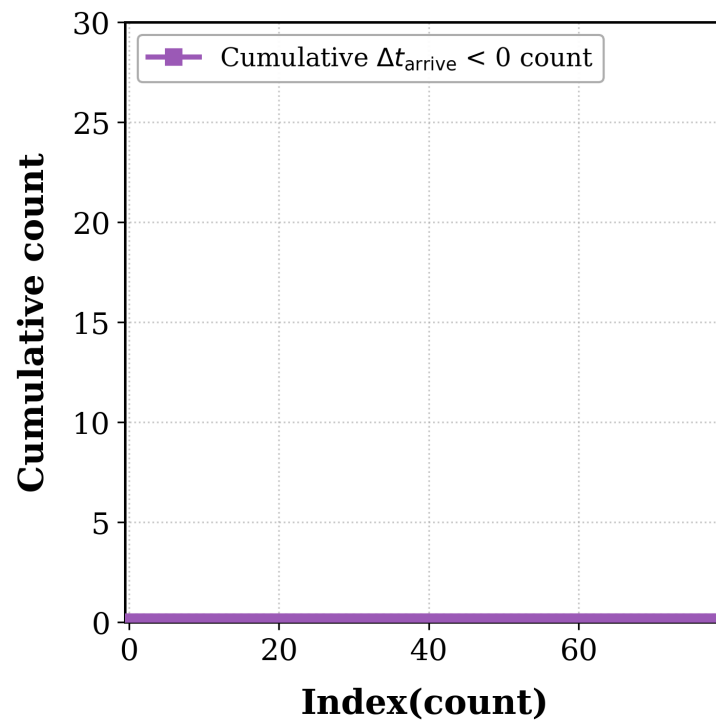


(b) With EWMA

Figure 4.2: EWMA processing validation and active adjustment for PNF arrival offset (Δt_{arrive}).



(c) Trend Observation



(d) Active Advance Adjustment

Figure 4.2: EWMA processing validation and active adjustment for PNF arrival offset (Δt_{arrive}) (cont.).

4.3 Scenario II: Performance Benchmarking

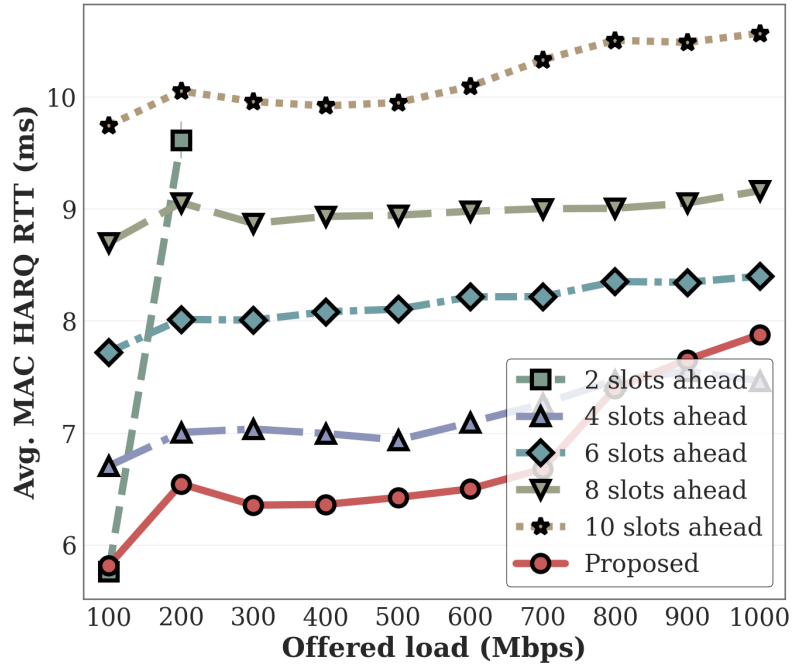
In this scenario, we evaluate the benchmark performance of the proposed adaptive timing mechanism against a standard static slots ahead configuration. Figure 4.3 presents the performance evaluation in terms of MAC layer latency and system throughput.

Figure 4.3 displays the performance evaluation under varying Offered Load. The proposed algorithm prioritizes ensuring optimal throughput, consistently maintaining optimal throughput performance under all load conditions. Simultaneously, while ensuring this optimal throughput, the algorithm maximizes the selection of the optimal slots ahead to optimize the MAC layer HARQ Round-Trip Time (RTT), effectively avoiding unnecessary extra overhead.

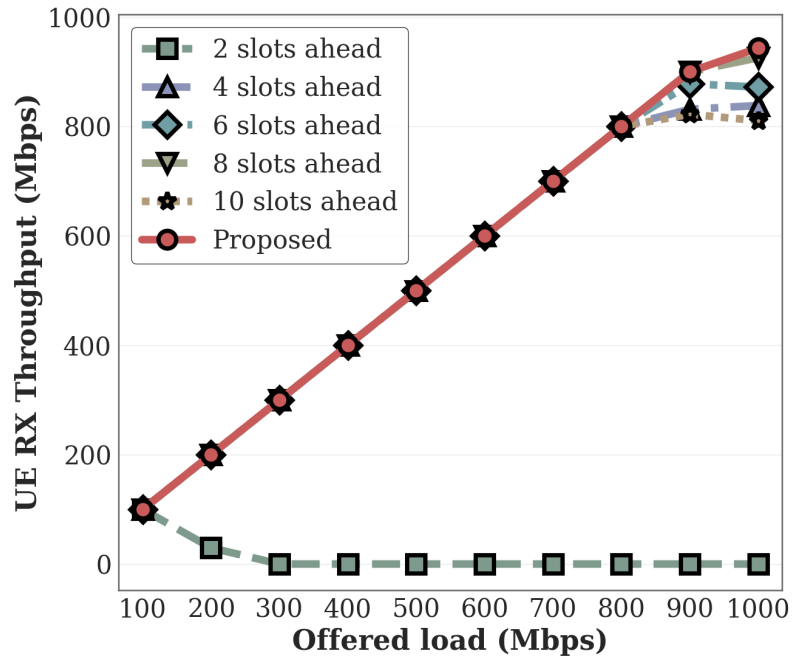


4.4 Scenario III: Stability under Network Congestion

To test the system's stability, we utilized Linux TC to inject artificial network congestion into the $T_2 - T_1$ and $T_5 - T_4$ segments, altering the One-Way Delay (OWD) and jitter. As shown in Figure 4.4a, when the delay exceeds the fixed timing window, the static configuration experiences a severe drop in throughput and frequent connection resets. In contrast, the proposed adaptive timing control dynamically expands the slots ahead (S_{ahead}). This expansion allows the MAC scheduler to trigger messages in advance, ensuring that messages can still arrive at the PNF within the valid



(a) Throughput Optimization

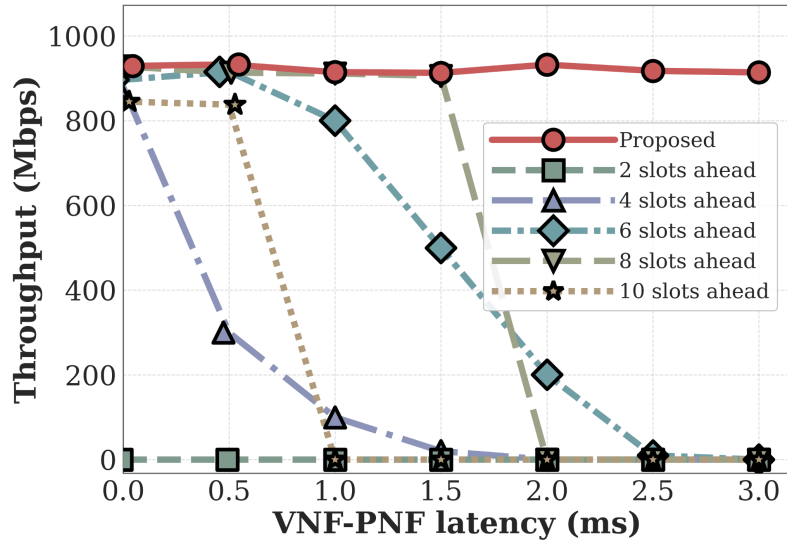


(b) MAC Latency and Reliability

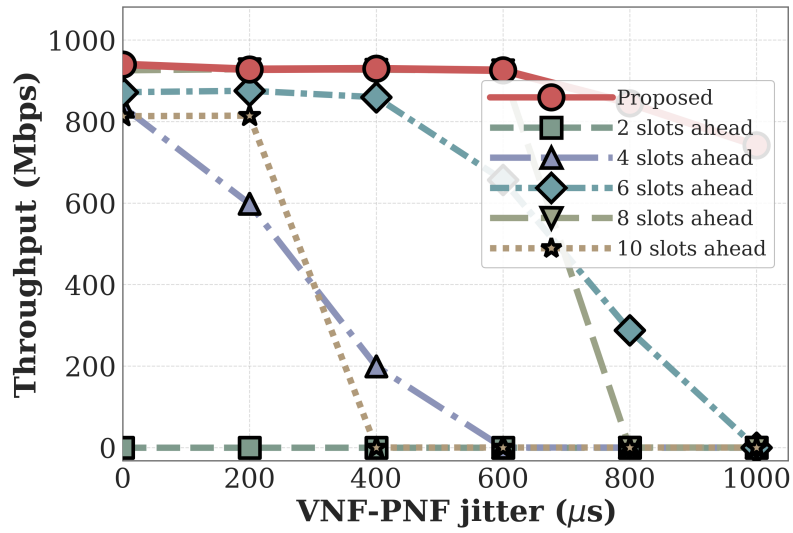
Figure 4.3: Performance Evaluation: MAC RTT Efficiency and System Throughput.

timing window despite the increased transmission delay. Therefore, even at congestion levels that would previously cause complete synchronization failure, the system can maintain stable throughput.

To further evaluate the impact of unpredictable network jitter between the VNF and PNF, we perform a second experiment using Linux TC to simulate random jitter on the fronthaul link. In this test, we run iPerf UDP traffic with an offered load of 1 Gbps to fully saturate the downlink transmission. We measure the resulting downlink (DL) receiver (RX) throughput at the UE, comparing different fixed slot-ahead configurations against our proposed adaptive timing mechanism. As shown in Figure 4.4b, when network jitter is high and unpredictable, fixed slot-ahead settings cannot handle the timing variations, leading to packet losses, sync failures, and a major drop in throughput. On the other hand, the proposed adaptive method dynamically adjusts the scheduling advance, keeping the transmission timing aligned. As a result, the UE achieves and maintains stable and high DL RX throughput even under severe jitter conditions.



(a) System Stability under Network Congestion



(b) Throughput Performance under Unpredictable Jitter

Figure 4.4: System stability and throughput performance under network congestion and jitter.

Chapter 5 Conclusion

This study addresses the transmission latency issues in the O-RAN Open Radio Access Network architecture by proposing an Adaptive Slots Ahead mechanism, with a specific focus on the Option 6 functional split architecture utilizing the nFAPI interface. In a non-ideal packet-switched network environment, traditional static slots ahead configurations cannot effectively handle unpredictable network jitter and server processing delays, easily leading to scheduling timeouts and synchronization failures. To overcome this challenge, this study designs a dynamic delay management algorithm based on the Exponentially Weighted Moving Average (EWMA) model. Through closed-loop feedback control, it continuously tracks and predicts end-to-end network transmission latency characteristics.

By integrating the OpenAirInterface (OAI) open-source software with commercial Pegatron O-RU equipment into an end-to-end physical verification platform, the experimental results demonstrate that the proposed delay management architecture significantly improves system stability. Under simulated network congestion scenarios, this mechanism actively compensates for timing offsets, ensuring that MAC layer scheduling commands accurately fall within the timing window of the physical layer, thereby completely eliminating connection drops caused by missing slot packets. Compared to traditional solutions that rely on expensive hardware acceleration or strict hardware-level PTP synchronization, the software-based solution proposed in this study demonstrates extremely high deployment flexibility and cost-effectiveness on general-purpose Commercial Off-The-Shelf (COTS) servers, providing a highly reliable key technological foundation

for the commercialization of future 5G/6G heterogeneous networks.



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